

Mohamed Aziz Skhiri

+216 52 237 532 | azizskhiri107@gmail.com | linkedin.com/in/mohamed-aziz-skhiri-498627254 | skhiri23.github.io



Profile | Data Scientist with hands-on experience building end-to-end ML pipelines, predictive models, and production systems. Reduced data processing time by 35% and built pipelines handling over 10,000 records/day. Full-stack aircraft fuel prediction platform (XGBoost, LSTM, RAG), deployed on the cloud, in partnership with Nouvelair Tunisie. Proficient in Python, SQL, Flask, React and AWS.

Key Skills

- Programming & Data:** Python, R, SQL, Bash, Git, Linux
- Machine Learning:** XGBoost, LSTM, Regression, Classification, Clustering, Hyperparameter Tuning, SHAP, Model Evaluation, TensorFlow/Keras, Scikit-learn
- Data Engineering & MLOps:** ETL Pipelines, REST APIs, Flask, FastAPI, Docker, AWS (EC2, S3), PostgreSQL, MySQL, MongoDB
- Visualization & BI:** Power BI, Tableau, Matplotlib, Seaborn, React/Recharts Dashboards

Professional Experience

- Data Scientist / Full-Stack Developer – Graduation Internship (PFE)** Feb 2026 – Jun 2026
Nouvelair Tunisie, IT & Projects Technology Department
 - Designed FuelTrack, an ML platform for monitoring, predicting and optimizing aircraft fuel consumption and CO₂ emissions, replacing an entirely manual workflow based on LIDO Flight, PDC FlightOps and OFP Excel files.
 - Developed and validated predictive models (XGBoost, Gradient Boosting, Random Forest, LSTM) to estimate Block Fuel, Trip Fuel, costs and CO₂, with per-registration bias correction (HFC) to improve accuracy.
 - Designed a RAG-based conversational assistant (Groq LLaMA 3.3-70B, with Gemini 2.5 Flash fallback) enabling natural-language querying of aggregated flight data in SQL.
 - Delivered a production-grade full-stack architecture (Flask, PostgreSQL, React) with JWT authentication and 4-tier role management, using a hybrid SCRUM + CRISP-DM methodology across 6 sprints, under academic and professional supervision.
- Full-Stack Developer Intern – Data/AI Module** May 2025 – Aug 2025
Elite Council Consulting
 - Built backend pipelines for an AI-driven attendance management system processing over 5,000 events/day.
 - Designed and deployed React dashboards for real-time analytics, improving reporting efficiency by 40%.
 - Refactored Flask/MongoDB modules, improving data flow reliability by 25%.
- Python Programming Instructor** Feb 2024 – Jun 2024
CEFI – Computer Training Center
 - Trained over 30 students in Python and data structures through hands-on analysis mini-projects; designed practical exercises and assignments on data manipulation.

Key Projects

- FuelTrack – Intelligent Aircraft Fuel Prediction & Recommendation Platform**
 - Prediction :** XGBoost, LSTM, Random Forest, Gradient Boosting models estimating Block Fuel, Trip Fuel and CO₂ (R² > 99%), with SHAP explainability and per-registration bias correction.
 - Route recommendation :** intelligent engine (Route Recommender & Route Decoder) suggesting the most fuel-efficient routes based on aircraft registration, with dynamic filters and real-time flight-count badges.
 - Dashboards & analytics :** Power BI dashboards and Python analytics (Pandas, Matplotlib/Seaborn) with anomaly detection (Z-score) and fuel savings tracking.
 - Chatbot :** RAG-based conversational assistant (Groq LLaMA 3.3-70B) enabling natural-language queries over flight data aggregated in SQL.
 - Production-grade full-stack architecture : Flask, PostgreSQL, React, JWT authentication and 4-tier RBAC.
- MedRAG – Drug Information RAG Assistant**
 - RAG system combining a **vector database** (embedding indexing, cosine similarity search) with an LLM to query drug leaflets : indications, precautions, interactions.
 - Ingestion and vectorization pipeline for drug leaflets, with optimized semantic retrieval, designed and deployed on Linux/Ubuntu.

Football Data Pipeline & Analytics Dashboard

- ETL pipeline processing over 20,000 matches with predictive feature engineering.
- **Flask + Docker API** exposing statistics, with interactive dashboards.

Education

Professional Master's in Data Science – Faculty of Sciences of Monastir (FSM) **2024–2026**

Relevant coursework: Machine Learning, Data Mining, Big Data Analytics, AI Systems

Graduation project: FuelTrack – Aircraft Fuel Monitoring & Optimization Platform (with Nouvelair Tunisie & ATDS)

Bachelor's in Computer Science – Software Engineering – Faculty of Sciences of Monastir (FSM) **2021–2024**

Graduation project: Human Resources Management System (HRMS)

Certifications

- Python for Data Science, AI & Development – IBM
- AWS Cloud Practitioner (Fundamentals) – AWS
- Full-Stack Web Development (React, Flask, MongoDB) – ECC
- Python Programming Certification – JNL Center

Languages

- **Arabic:** Native language
- **English:** Professional level
- **French:** Intermediate level